EESC3600 Ecosystems and Climate (2026 Spring)

Topic 1 - Ecosystems, Global Plant Biogeography and Climate

- **Ecosystem**: a community of interacting organisms (plants, animals, fungi, microbes) in a habitat together with their physical environment (air, water, soil, etc.).
- **Biosphere**: the global sum of all ecosystems.
- **Biogeophysical effects**: terrestrial vegetation regulates the surface energy balance in the form of radiation, sensible heat (conduction and convection), and latent heat (evapotranspiration), affecting local climate and environment.
  - ***Key land-atmosphere biogeophysical processes***
    1. ***Albedo*** (α): the fraction of incoming solar radiation that is reflected to space by a surface. Surfaces that are light-colored and reflective (like fresh snow, ice, deserts, or sparse shrubland) reflect most sunlight.
       Physical rationale:
       Different surfaces have different physical properties that determine how they interact with solar radiation. Surfaces that are dark and absorbent (like forests, open ocean, or wet soil) absorb most sunlight. Lower albedo indicates higher absorption of solar radiation and causes a warming effect on the surface.
    2. ***Surface roughness***: a measure of the aerodynamic resistance to wind flow. It is determined by the height, shape, and spacing of surface elements (like trees, crops, or grass). Often quantified by the aerodynamic conductance (g_a).
       Physical rationale:
       Rough surfaces (like a forest) create physical obstructions that generate turbulence in the air layer above them. Smooth surfaces (like a lawn, bare soil, or shrubland) allow wind to pass more smoothly (laminar flow). The more turbulence, the higher the aerodynamic conductance, meaning that heat and water vapor can be mixed and transported away from the surface more efficiently. In terms of flux, higher g_a allows the surface to dissipate greater flux of energy (sensible heat flux H, latent heat flus λE) away.
    3. ***Canopy physiology / Stomatal conductance*** (g_s): the physiological control of water vapor loss (transpiration) and CO_2 uptake (photosynthesis) by plants, primarily through the opening and closing of stomata.
       Physical rationale:
       Stomata are pores that open to allow CO_2 in for photosynthesis, but this inevitably lets water vapor out (transpiration). Plants optimize g_s based on environmental conditions (light, temperature, humidity, soil moisture) to balance carbon gain and water loss. Transpiration is a major component of evapotranspiration (ET), the rate of transpiration is directly controlled by g_s and the vapor pressure deficit D. Higher g_s allows the higher rate of transpiration that causes a higher latent heat flux (λE), cooling the surface.
    4. ***Soil moisture***: the amount of water contained in the soil acting as the water supply for plants and as a direct source for evaporation. It creates strong climate feedback. For example, a lack of rain leads to dry soil, which reduces evaporative cooling, leading to higher temperatures, which can exacerbate drought conditions.
       Physical rationale:
       Water has a high specific heat capacity and is required for evaporation. The amount of moisture in the soil determines: (1) evaporative cooling; (2) plant stress determining g_s. Soil moisture is the ultimate control on whether the surface energy is used for heating the air (sensible heat) or evaporating water (latent heat).
- **Biogeochemical effects**: Life modifies climate and the environment (air, water, soils) via various physiological and biochemical processes, which form a complex network of pathways that allows the cycling of chemical elements.
  - ***Key land-atmosphere biogeochemical processes***
    1. ***Carbon cycle***
       Core processes: photosynthesis, respiration & decomposition, disturbances.
       Feedbacks:
       Terrestrial CO_2 fertilization feedback: rising atmospheric CO_2 can stimulate photosynthesis (A_n increases), creating negative feedback (land sink gets larger, dampening CO_2 rise). However, this effect may be limited by nutrients (nitrogen, phosphorus) and water.
       Temperature-decomposition feedback: warming temperatures accelerate microbial decomposition (k increases), especially in high-latitude soils (boreal forests, permafrost). This releases more CO_2 and CH_4, creating a positive feedback that amplifies warming.
       Ocean CO_2 uptake: the ocean absorbs CO_2 via the solubility pump. This uptake is limited by carbonate ion (CO_3^{2-}) availability and ocean warming. As the ocean warms and acidifies, its ability to absorb CO_2 decreases, a positive feedback to climate change.
    2. ***Reactive nitrogen cycle***: transformation of nitrogen between its inert atmospheric form (N_2) and its reactive forms.
       Core processes:
       Nitrogen fixation: conversion of inert N_2 gas into organic nitrogen (or ammonia) by lightning or specialized microbes (e.g., in legume root nodules). This is the natural source of "new" nitrogen to ecosystems.
       Nitrification & Denitrification: Microbial processes in soils and water.

Nitrification: ammonium (NH_4^+) is converted to nitrate (NO_3^-). A byproduct of this process is nitrous oxide (N_2O).

Denitrification: under anaerobic (low-oxygen) conditions, microbes use nitrate (NO_3^-) as an electron acceptor, converting it back to N_2 gas. A byproduct of this process is also N_2O, as well as nitric oxide (NO).

Biomass burning & fossil fuel combustion: high-temperature combustion oxidizes organic nitrogen and atmospheric N_2, producing nitrogen oxides (NO_x=NO+NO_2).

Atmospheric impacts:

N_2O (nitrous oxide): a long-lived greenhouse gas (~300x more potent than CO_2 over 100 years). It also destroys stratospheric ozone.

NO_x (nitrogen oxides): catalyzes the formation of tropospheric ozone (O_3), which is a potent GHG and air pollutant.

NH_3 (ammonia): reacts with nitric and sulfuric acid in the atmosphere to form secondary inorganic aerosols (e.g., ammonium nitrate, ammonium sulfate), which are a major component of PM2.5 (air pollution) and affect radiative forcing.

Deposition (acid Rain & eutrophication): deposition to the surface will cause acidification (acid rain) and eutrophication (nutrient over-enrichment) of ecosystems.

3. ***Aerosols***: tiny solid or liquid particles suspended in the atmosphere.
   Core Sources (Biotic & Abiotic):
   Primary biogenic aerosols: Direct emission of particles. Examples: pollen, spores, plant fragments, sea spray (from wave action, contains organic matter), mineral dust (from dry, sparsely vegetated areas like deserts. Vegetation loss increases dust emission).
   Secondary organic aerosols (SOA): Gases emitted from the biosphere (especially BVOCs) that oxidize in the atmosphere and condense to form particles. Example: biomass burning aerosols (smoke) from wildfires and agricultural burning contains black carbon (BC) and organic carbon (OC).
   Climatic Impacts:
   Direct effect (radiative forcing)
   Black carbon (BC): Strongly absorbs solar radiation, warming the atmosphere and dimming the surface.
   Organic carbon (OC) / Sulfate / Nitrate: Scatters solar radiation back to space, cooling the planet.
   Indirect effect (cloud albedo & lifetime)
   Aerosols act as cloud condensation nuclei (CCN). More aerosols lead to more, but smaller, cloud droplets. This makes clouds brighter (higher albedo, cooling) and can suppress precipitation, increasing cloud lifetime (cooling).
4. ***Biogenic volatile organic compounds***: carbon-based gases emitted by plants that are highly reactive in the atmosphere.
   Core Processes:
   Synthesis & emission: plants synthesize BVOCs for various reasons: defense against herbivores/pathogens, communication with other plants, protection from heat/oxidative stress, or as byproducts of metabolism.
   Key compounds: The most important are isoprene and monoterpenes. Emission rates are strongly dependent on temperature and light. High temperatures and high light generally increase BVOC emissions.
   The Climatic & Atmospheric Impacts:
   Tropospheric ozone (O_3) formation: BVOCs react with nitrogen oxides to form ozone in the presence of sunlight.
   Secondary organic aerosol (SOA) formation: BVOCs oxidize in the atmosphere, and their oxidation products can condense to form SOA as a major source of aerosols.
   Methane lifetime: BVOC chemistry can affect the concentration of the hydroxyl radical (OH), which is the primary "detergent" that removes methane (CH_4) from the atmosphere. By altering OH levels, BVOCs can indirectly influence the lifetime and concentration of this potent GHG.
   Urban greening caution: in urban areas with high NO_x (from vehicles), planting high-BVOC trees will exacerbate ozone pollution, negating some of the cooling benefits of greening.

- **Descriptive methods for an ecosystem**
  - **Community composition**: the identity and abundance of different species present in a community.
    Ecological principles: (1) Succession: Community composition changes dramatically over time after a disturbance. Early successional communities are composed of different species (fast-growing, high light requirement) than late successional communities (slow-growing, shade-tolerant). (2) Disturbance & Stress: Changes in climate (temperature, precipitation) or disturbance regimes (fire, pests) will alter community composition. Species that are better adapted to the new conditions will become more abundant, while others may decline or disappear.
  - **Population size structure**: the distribution of individuals in a population among different size classes or age classes. For plants, size (e.g., diameter at breast height, height) is often a better indicator of age and ecological role than chronological age.
    Ecological principles: (1) Forest Succession & Carbon Cycling: Size structure influences the current and future carbon storage of a forest. A forest with many large trees (a maturing forest) has a large biomass pool. The size structure also dictates which trees are going to die and become litter in the near future, influencing the litter

pool and decomposition rates; (2) Response to Disturbance: A fire or a pest outbreak that kills large trees will radically alter the size structure, shifting it from a bell-shaped curve back to an inverse J-shape as the forest regrows (secondary succession).

- **Population distribution**: how individuals of a population are arranged in space within their habitat. Three main patterns: (1) randomly distributed, resources are uniformly abundant and there is no strong interaction (e.g. dandelions in a uniformly grassy field), (2) uniform, individuals are evenly spaced, which indicates intense competition for resources (e.g. water in a desert, or light in a dense forest) or territorial behavior (e.g. allelopathy, where plants release chemicals to inhibit neighbors), (3) clumped, individuals occur in groups, as resources are patchy, reproduction is local and it provides protection from predators or climate.
  Ecological principles: Uniform distribution is a direct manifestation of competition. Spatial patterns are a result of how seeds and pollen are dispersed. Wind-dispersed seeds might lead to a more random pattern over a large area, while animal-dispersed seeds might be clumped under perching trees. Clumped distribution acknowledges that the environment is not uniform. This connects to edaphic factors (soil type, depth) mentioned as a limitation of pure biogeographical models.
- **Ecosystem structure**: the properties and quantities of materials/energy stored in different compartments, known as pools or reservoirs.
- **Ecosystem functions**: the processes governing the movement, or flows, of materials/energy between pools. Each flow has a transfer rate dependent on environmental conditions.
- **Turnover Time**: The time it takes for a material to be completely replaced in a pool, determined by the pool size and the flow rate.

- **Ecosystem carbon flow and storage**
  Gross primary production (GPP): GPP = gross rate of carbon uptake by photosynthesis (usually in $\mathrm{gC\ m^{-2}\ yr^{-1}}$)
  Net primary production (NPP): NPP = GPP – autotrophic respiration (by plants)
  Net ecosystem production (NEP): NEP = NPP – heterotrophic respiration (by all other organisms)
  Net biome production (NBP): NBP = NEP – disturbances (e.g. wildfires, harvesting, land use, insect outbreaks)
  Different forest types (tropical, temperate, boreal) have vastly different magnitudes of carbon fluxes.
  - ***Forest types and carbon fluxes***
    1. **Tropical humid evergreen forest** (e.g., Amazon, Congo)
       Very large fluxes. GPP is extremely high (e.g., 3000+ g C m^{-2} yr^{-1}). Autotrophic and heterotrophic respiration are also very high. NPP is high, but a significant portion of GPP is respired.
       Ecological & physical rationales:
       Climate: high year-round temperature and precipitation. No thermal or water limitation.
       Physiology: optimal conditions for enzyme kinetics (Rubisco, etc.). High leaf area index (LAI) means the canopy intercepts a huge amount of PAR (photosynthetically active radiation), leading to very high gross primary production (GPP).
       Biogeochemistry: high temperatures and moisture lead to very rapid decomposition (high k value). This means: Fast Nutrient Cycling: Litter decomposes quickly, releasing nutrients that are immediately taken up by plants, fueling high GPP.
       High respiration: the high rate of decomposition means heterotrophic respiration (R_h) is a very large flux, almost as large as autotrophic respiration (R_a).
       Small carbon pools (in soil): because decomposition is so fast, carbon doesn't build up in the soil. The massive carbon pool is in the living biomass (the trees themselves).
    2. **Boreal semiarid evergreen forest** (e.g., Siberia, Canada)
       Small Arrows. GPP is low (e.g., 400-800 g C m^{-2} yr^{-1}). Respiration fluxes are also low.
       Ecological & physical rationales:
       Climate: the dominant limitation is low temperature and a very short growing season. Water is also less available (much of it is frozen).
       Physiology: low temperatures slow down all biochemical reactions, including photosynthesis. Low solar angle and short days limit PAR. Coniferous needles have a lower photosynthetic capacity than broadleaves.
       Biogeochemistry: this is the key part. Cold temperatures and waterlogging (from permafrost) severely inhibit microbial activity. This means: Very Slow Decomposition (low k): Litter production is low, but decomposition is even slower. Nutrient Limitation (High C:N): Because decomposition is slow, nutrients (especially nitrogen) remain locked up in the litter and soil organic matter. This creates a strong nitrogen limitation on plant growth (Principle of Limiting Factors), which keeps NPP low. Large Carbon Pools (in soil): Paradoxically, despite low productivity, boreal forests store a huge amount of carbon. Because decomposition is so slow, organic matter (litter, humus) accumulates over millennia, creating a massive soil carbon pool. The arrows (fluxes) are small, but the soil pool box is enormous.
    3. **Temperate forests** (Deciduous/Evergreen) (e.g., Eastern North America, Central Europe)
       Fluxes are intermediate between tropical and boreal. GPP is high during the growing season but drops to near zero in winter (for deciduous).
       Climate: clear seasonality, temperature is the main driver of the cycle. Warm, moist summers allow for high productivity, while cold winters induce dormancy.

Physiology: deciduous trees have high photosynthetic rates during the leaf-on period but cannot photosynthesize when leafless. Evergreen temperate trees can photosynthesize on warmer winter days but at a low rate.

Biogeochemistry: moderate temperatures and moisture allow for moderate decomposition rates. Nutrient cycling is faster than in boreal forests but slower than in the tropics. Soil carbon pools are moderate.

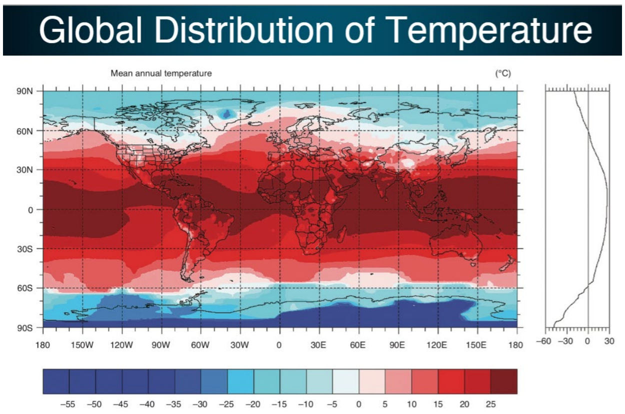
Carbon is stored in different pools (living biomass, standing dead, forest floor, mineral soil).
The distribution among pools varies significantly by forest type (e.g., aspen vs. black spruce).
<u>Ecological & physical rationales:</u>

- 1. Quaking Aspen (Deciduous Broadleaf): relatively open, fast-growing and shade-intolerant. They invest heavily in aboveground growth (stems and leaves) to compete for light. It allocates carbon to stems (wood) and leaves, decomposes relatively quickly.
- 2. Black Spruce (Needleleaf Evergreen): dense, closed-canopy, slow-growing, shade-tolerant, and can survive in nutrient-poor, cold, water-logged conditions. They have a conical shape to shed snow. Despite slow growth, the trees are densely packed and can be old. They store a significant amount of carbon in wood, but their productivity (NPP) is lower than aspen. Its growth is limited by cold soils and lack of nutrients. The cold, dry conditions slow the decomposition of wood. There is a massive, thick layer of partially decomposed spruce needles and mosses. The organic layer and underlying permafrost impede drainage, creating waterlogged, anaerobic conditions that further inhibit decomposition.

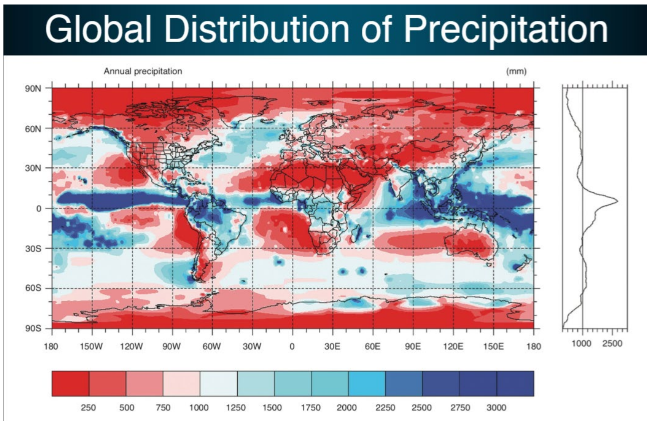
- **Potential limitations to vegetation production**

Basic needs of living organisms: an energy source, water, nutrients (C, N, P, etc.), suitable temperature, etc.
Limitations to NPP: global vegetation production is primarily limited by three factors:

1. **Solar Radiation**: fundamental limit on the energy available for photosynthesis.
2. **Water Balance**: the availability of water (precipitation minus losses) is crucial for plant physiology and transport.
3. **Temperature**: controls the rate of biochemical reactions.



Latitudinal distribution corresponds mostly with incoming solar radiation.

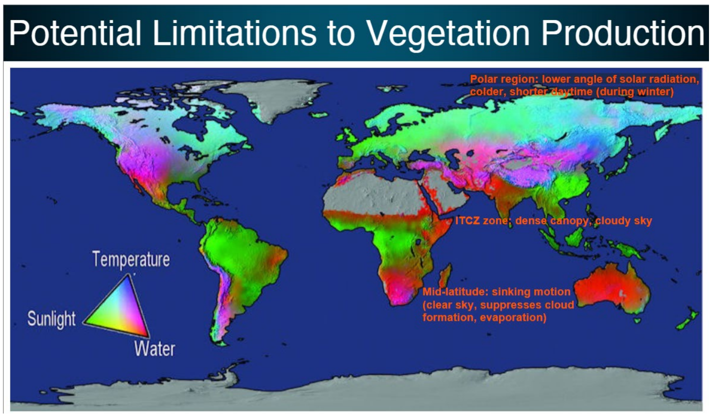


Equatorial regions receive the greatest amount of rainfall.
Subtropical/temperate deserts and polar regions receive the least.

- **Evapotranspiration** (ET): the sum of evaporation from soil/water and transpiration from plants. It is an integrated measure of available energy (temperature) and water (precipitation). ET corresponds very closely with NPP. ET depends on available energy (solar radiation, temperature), air humidity, wind speed, soil type and moisture, and plant physiology.
- **Global plant biogeography**

World's major climate zones are classified by systems like Köppen, based on temperature and precipitation patterns.

- Geographic pattern of natural vegetation (biomes like tropical rainforest, desert, tundra) closely matches these climate zones, which is largely shaped by the distribution of temperature and precipitation.
- **The global map of NPP limitations**

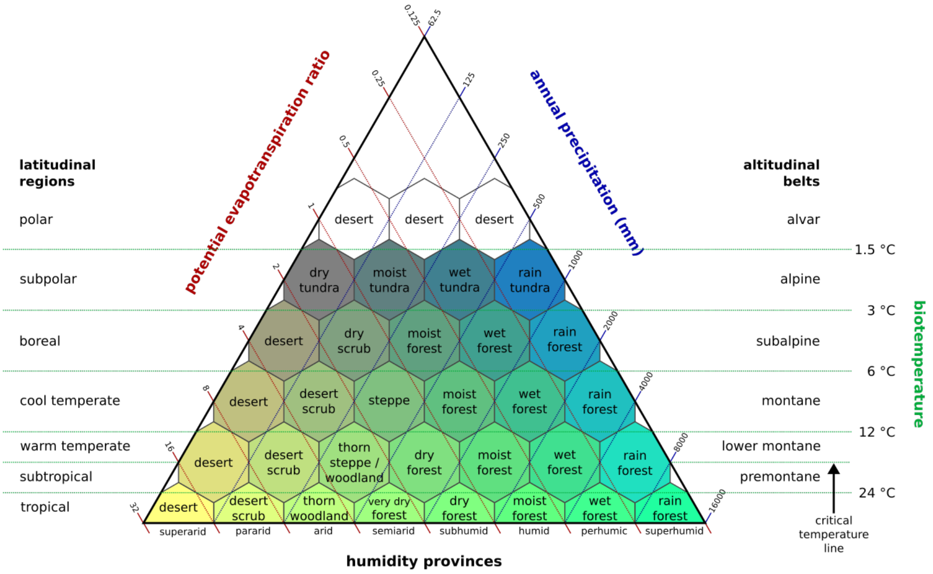


- ◆ **The Intertropical Convergence Zone (ITCZ)**: light-limited in equatorial regions (e.g., Amazon Basin, Congo Basin, Maritime Southeast Asia). The trade winds from the Northern and Southern Hemispheres meet here, warm, moist air rises, cools, and condenses. This rising motion creates a band of persistent, deep convective clouds. The vegetation itself also contributes to high evapotranspiration, which fuels more clouds. The result is persistent, thick cloud cover. Within the forest itself, the dense canopy (high LAI) means that leaves deep in the understory are severely light-limited.

- ◆ **The subtropics**: water-limited around 20°-30° north and south (e.g., Sahara Desert, Arabian Peninsula, Australian Outback, Sonoran Desert). This is the descending branch of the Hadley Cell. As air sinks, it warms adiabatically and dries out, suppressing cloud formation. The intense solar radiation at the surface, combined with dry air, leads to very high potential evapotranspiration.
- ◆ **The mid-latitudes**: temperature and water-limited at 30°-60° north and south (e.g., Europe, Eastern North America, East Asia). This region is influenced by the Ferrel Cell and the westerlies, characterized by migrating cyclones and strong seasonality. In winter, cold temperatures make photosynthesis impossible. In summer, warm temperatures and adequate moisture mean radiation (light) is the primary driver, but brief summer droughts can impose temporary water limitation. The growing season is therefore constrained by both the duration of frost-free conditions (temperature) and the availability of soil moisture during the warm months (water).
- ◆ **The polar regions**: temperature and light-limited at >60° north and south (e.g., Arctic Tundra, Boreal Forests/Taiga, Antarctica). The solar angle is low year-round, spreading energy over a larger surface area, and winters bring extended periods of darkness (polar night). Low temperatures directly slow all biochemical reactions.

- ■ **Influence of climate on vegetation structure**: leaf area index and canopy height generally increase with annual precipitation.
- ■ **Influence of climate on plant productivity**: elevated CO_2 (fertilization effect), rising temperature (increases metabolic activities and increasing water loss), and sufficient water availability can increase plant productivity.
- ■ **Paleoecology**: past vegetation and climate can be reconstructed using proxies, e.g. pollen.
- ■ **Biogeographical approach for predicting future climate change**: by mapping the future climate onto a predefined set of biomes, predicting how vegetation distribution will shift with climate change.

- **Holdridge vegetation classification**



potential ET ratio = annual potential ET / annual precipitation

- **"Top-down" biogeographical model and limitations**

"Top-down": focus on the large-scale relationship from climate to vegetation.

1. **Unidirectional assumption**: one-way relationship from climate to vegetation.
   In reality, the relationship is bidirectional. Vegetation changes also affect climate through biogeophysical (albedo, ET) and biogeochemical (carbon cycle) feedback. Climate and vegetation coevolve.
2. **Equilibrium assumption**: vegetation is always in equilibrium with climate.
   If climate changes, the model assumes vegetation instantly jumps to its new prescribed state, ignoring transient dynamics and the time lags involved in plant migration, growth, and community assembly.
3. **Ignores physiological adaptation**: plants can acclimate or adapt physiologically to new conditions (e.g., tundra species evolving to survive warmer temperatures). This creates plasticity not captured by a rigid biome classification.
4. Other factors:
   i. **CO_2 fertilization**: the direct effect of elevated CO_2 on plant physiology (water-use efficiency, growth rates).
   ii. **Disturbance regimes**: changes in fire frequency, pest outbreaks, and other disturbances that can drastically alter vegetation patterns.
   iii. **Edaphic factors**: local soil type, depth, and chemistry can override climatic controls on where plants can grow.
   iv. **Human land use**: direct modification of the landscape by humans (agriculture, urbanization) is a major factor not included.

Topic 2 - Plant and Canopy Ecophysiology: Photosynthesis and Transpiration

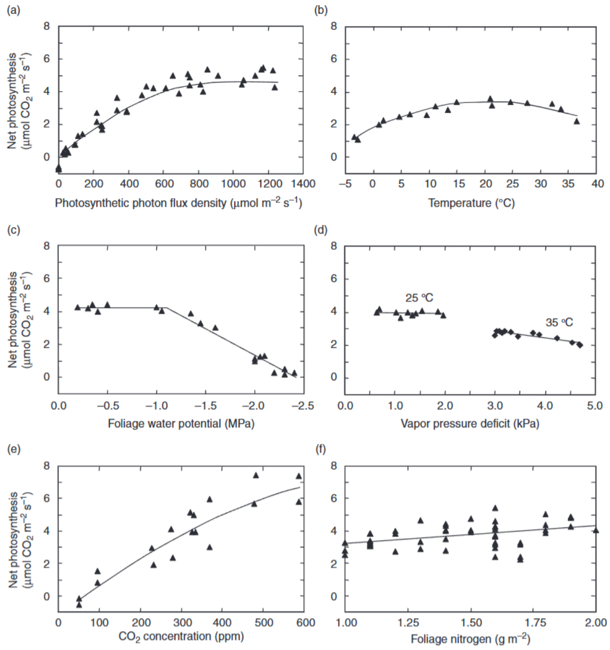
- **Photosynthesis**: the process by which autotrophs (including photosynthetic land plants and marine phytoplankton) fix atmospheric CO_2.
  $$\mathrm{CO_2 + H_2O + light\ energy \rightarrow CH_2O + O_2}$$
  Redox: Carbon is reduced (oxidation state from +4 in CO_2 to 0 in CH_2O).
  Key enzyme: Ribulose-1,5-bisphosphate carboxylase/oxygenase (RuBisCO).
  Product: Energy from sunlight is converted into chemical energy, stored in the chemical bonds of reduced carbon compounds (carbohydrates, lipids).
  Relation to the climate: net cooling effect, as $\mathrm{CO_2}$ is absorbed.
- **Respiration and decomposition**: the process of oxidizing reduced carbon to release energy.
  $$\mathrm{CH_2O + O_2 \rightarrow CO_2 + H_2O + energy}$$
  Autotrophic respiration: respiration by plants themselves.
  Heterotrophic respiration: respiration by herbivores, predators, scavengers, detritivores, and decomposers (fungi, and microbes) that consume organic matter.
  Decomposition returns carbon and nutrients bound in dead organic matter back to the environment.
- **The machinery of photosynthesis**
  Photosynthesis takes place in specialized organelles called chloroplasts.
  Light reaction occurs on the thylakoid membrane.
  1. Light excites electrons in Photosystem II (PSII/P680).
  2. Lost electrons are replaced by splitting H_2O, producing O_2 and releasing H^+ into the thylakoid lumen.
  3. The excited electrons go through an electron transport chain (ETC) from PSII to PSI (P700), and along the way energy is released to pump H^+ from the stroma into thylakoid lumen, building a proton gradient across the thylakoid membrane.
  4. The electrons received by PSI are excited again by the absorption of photons and finally used to reduce $NADP^+$ to NADPH.
  5. As protons flow down the gradient back to the stroma via the ATP synthase complex, electrochemical energy is released to synthesize ATP from ADP in the photophosphorylation process.

  Photons from sunlight are absorbed primarily by the chlorophyll pigment. Accessory pigments such as carotene help absorb a wider spectrum of sunlight.
  A cluster of pigment molecules together form a photosystem (PS).
  Plants can only make use of radiation of wavelengths between 0.4–0.7 μm.
  Dark reaction occurs in the stroma of the chloroplast.
  C3 pathway:
  1. Enzyme RuBisCO combines $\mathrm{CO_2}$ with RuBP (a 5-carbon compound).
  2. This forms two molecules of 3-PGA (a 3-carbon compound).
  3. ATP and NADPH from the light reaction are used to convert 3-PGA into G3P (GAP).
  4. G3P is used to produce carbohydrates (like glucose) and to regenerate RuBP.

  Overall: ATP and NADPH are used, organic compounds are generated through
  $$n\mathrm{CO_2} + 2n\mathrm{H_2O} + h\nu \rightarrow (\mathrm{CH_2O})_n + n\mathrm{O_2} + n\mathrm{H_2O}$$
  The problem with C3: RuBisCO can also fix O_2 instead of $\mathrm{CO_2}$, a process called photorespiration. This wastes energy and reduces net carbon uptake, especially in hot, dry, and high-light conditions.
  The C4 Solution: a spatial separation that concentrates $\mathrm{CO_2}$ at the RuBisCO site, minimizing photorespiration, increasing net photosynthetic rate and improves water use efficiency, especially at high light levels and warm temperatures. Another important carboxylating enzyme: PEP carboxylase.
  C4 mechanism:
  1. In mesophyll cells, $\mathrm{CO_2}$ is fixed into a 4-carbon compound (hence C4).
  2. This compound is transported to bundle sheath cells.
  3. It releases $\mathrm{CO_2}$ for the Calvin Cycle, creating a high $\mathrm{CO_2}{:}\,\mathrm{O_2}$ ratio.

  High light levels → high O2 from light reaction → high photorespiration → low photosynthetic rate → needs C4 pathway to reduce photorespiration
  Warmer → more transpiration → potential water stress → requires C4 pathway with high water use efficiency
  Example of C3 plants: trees, wheat, rice
  Example of C4 plants: grasses, corn, tropical savanna plants
- **Light reflection and vegetation indices**
  Spectral reflectance: green leaves absorb >85% of visible light (photosynthetically activeradiation, PAR) for photosynthesis. They reflect more than half of the near-infrared radiation to avoid overheating.
  Normalized difference vegetation index (NDVI): a measure of greenness based on the difference in reflectance.
  $$\mathrm{NDVI} = (r_{\mathrm{nir}} - r_{\mathrm{red}})/(r_{\mathrm{nir}} + r_{\mathrm{red}})$$
  r_{nir}: reflectance in near-infrared; r_{red}: reflectance in red wavelengths. High values (close to 1) indicate dense, healthy vegetation.

- **Environmental controls and limitations of photosynthesis**
  Photosynthetic rate is dynamically controlled by a variety of environmental factors and is limited by the slowest of several metabolic processes.



  1. Light (PPFD - Photosynthetic Photon Flux Density):
     The initial slope of the light response curve shows that photosynthesis is light-limited. In this phase, more light leads to more ATP and NADPH production.
     A plateau is reached where photosynthesis is co-limited by other factors (like CO_2 or enzyme activity). At this point, chlorophyll and enzymes are fully activated.
  2. Intercellular $\mathrm{CO_2}$ Concentration (c_i):
     $\mathrm{CO_2}$ is the substrate for the Calvin Cycle. Low c_i (often due to stomatal closure) directly limits the Rubisco-limited rate.
  3. Temperature:
     Low Temperatures: Slow down the enzymatic reactions of the Calvin Cycle.
     Optimal Range: There is an optimum temperature range where enzyme activity is maximized.
     High Temperatures (Heat Stress): Cause a sharp decline in photosynthesis due to:
     - Denaturation of enzymes (like Rubisco activase).
     - Impairment of electron transport in the light-dependent reactions.
     - Destabilization and destruction of cell membranes.
  4. Water Availability (Soil Water):
     Decreasing soil water content leads to a decrease in leaf water potential and an increase in the vapor pressure deficit (D). This triggers stomatal closure (lowers g_s) to prevent water loss, which in turn reduces c_i (the $\mathrm{CO_2}$ supply) and limits photosynthesis.
  5. Nutrients (e.g., nitrogen):
     Nitrogen is a critical component of Rubisco and chlorophyll. Nitrogen deficiency directly limits the plant's capacity to build and maintain the photosynthetic machinery.
- Modeling photosynthesis (Farquhar model)
  Photosynthesis rate (A) is limited by the slowest of three processes:
  1. **Rubisco-limited rate (A_c):** the rate at which RuBisCO can fix $\mathrm{CO_2}$.
     Limiting factors:
     - Availability of $\mathrm{CO_2}$ at the Rubisco site (depends on stomatal opening and c_i).
     - The kinetic properties (affinity for $\mathrm{CO_2}$ vs. $\mathrm{O_2}$) and the amount of Rubisco enzyme present (which is linked to nitrogen availability).

$$A_c = \frac{V_{\text{cmax}}(c_i - \Gamma_*)}{c_i + K_c(1 + o_i/K_o)}$$

     V_{cmax}: the maximum carboxylation rate ($\mathrm{\mu mol\ m^{-2}\ s^{-1}}$)
     c_i: intercellular $\mathrm{CO_2}$ concentration ($\mathrm{\mu mol\ mol^{-1}}$)
     Γ_*: $\mathrm{CO_2}$ compensation point where photosynthesis balances leaf respiration ($\mathrm{\mu mol\ mol^{-1}}$)
     K_c: M-M constant for carboxylation ($\mathrm{\mu mol\ mol^{-1}}$)
     K_o: M-M constant for oxygenation ($\mathrm{\mu mol\ mol^{-1}}$)
     o_i: $\mathrm{O_2}$ concentration=209 $\mathrm{mmol\ mol^{-1}}$
  2. **RuBP-limited rate / Light-limited rate (A_j):** the rate at which RuBP, the $\mathrm{CO_2}$ acceptor, can be regenerated.
     Limiting factor: fundamentally controlled by the availability of photons (light) to produce ATP and NADPH.

$$A_j = \frac{J(c_i - \Gamma_*)}{4(c_i + 2\Gamma_*)}, \Theta_J J^2 - (I_{\text{PSII}} + J_{\max})J + I_{\text{PSII}} J_{\max} = 0, I_{\text{PSII}} = 0.5\Phi_{\text{PSII}}\alpha_l I$$

     Depends on the electron transport rate (J), which is a function of PAR incident on a leaf and various leaf parameters such as leaf absorptance, quantum yield and maximum potential electron transport rate.
  3. **Product-limited rate (A_p):** the rate at which the products of photosynthesis (G3P) can be utilized.
     Limiting Factor: This is a sink limitation. If sugars aren't being used for growth or export, they accumulate. This accumulation signals the chloroplast to slow down, partly because the phosphate needed to make ATP gets "tied up" in the phosphorylated sugars and isn't released back into the chloroplast. This is a rare limitation under most natural conditions but can occur.

$$A_p = 3T_p$$

  Net photosynthesis (A_n):

  $A_n = \min(A_c, A_j, A_p) - R_d$ where R_d is the leaf mitochondrial respiration rate.

- **Fluxes for surface-atmosphere exchange**

$$F = cg(\varphi_s - \varphi_a), r = 1/g$$

  - Stomata respond to multiple environmental factors to optimize $\mathrm{CO_2}$ gain against water loss.
    Jarvis's empirical model

    $$g_s = g_s(I)f_1(T)f_2(D)f_3(\psi)f_4(c_a)$$

    Photosynthetic photon flux density (I): g_s increases with light.
    Temperature (T): Has an optimal range.
    Vapor pressure deficit (D): g_s decreases as D increases (to save water).
    Foliage water potential (ψ): g_s decreases as water potential becomes more negative (water stress).
    Ambient CO_2 (c_a): g_s decreases as c_a increases.
  - **Resistance model for land-atmospheric exchange**

    $$r = r_a + r_b + r_c$$

    r_a: aerodynamic resistance (turbulent diffusivity, wind, stability, roughness, etc.)
    r_b: boundary-layer resistance (viscosity, molecular diffusivity, etc.)
    r_c: surface/canopy resistance (surface characteristics)
  - **Heat and gas exchange in leaves**
    Stomata: pores or openings on leaves that allow the exchange of gases between plants and the atmosphere.
    In: CO_2, O_3, SO_2, etc.
    Out: H_2O, BVOCs (biogenic volatile organic compounds)
    Stomata open to allow CO_2 uptake during photosynthesis, but close to prevent excess water loss during transpiration.
    Stomatal conductance (g_s): a measure of how open the stomata are.
    It is common to treat the conductances for heat (e.g., g_{bh}) and for water vapor (e.g., g_{bw}) to be the same. The reason is that movement of both heat energy and water vapor through that stagnant air layer is governed by the same laws of molecular diffusion, and the minor theoretical differences are smaller than the practical uncertainties of field measurements of the boundary layer. Heat is mainly transported by water vapor in the boundary layer.
    Stomata are regulated to maximize CO_2 gain via photosynthesis and minimize water loss via transpiration. A_n and g_s vary almost linearly for a given set of conditions. Plants achieve this optimization by adjusting g_s to maintain a nearly constant c_i/c_a ratio for given environmental conditions, thus, assuming $g_b \gg g_s$:

    $$g_s \approx \frac{1.6\ A_n}{c_a(1 - c_i/c_a)}$$

    $c_i/c_a \approx 0.65 - 0.80$ for C3 plants; $c_i/c_a \approx 0.40 - 0.60$ for C4 plants
    Ball-Berry stomatal conductance model: as a diffusion process modulated by both boundary-layer conductance (g_b) and stomatal conductance (g_s). Sensible heat exchange (heat transfer by conduction and convection) is controlled by g_b alone.

    $$g_s = g_0 + g_1(A_n h_s/c_s)$$

    h_s: fractional humidity at leaf surface (unitless)
    c_s: leaf surface CO_2 concentration (μmol mol−1)
    g_0: minimum conductance ≈ 0.01 $\mathrm{mol\ H_2O\ m^{-2}\ s^{-1}}$ for C3 plants
    g_1: slope of relationship ≈ 9 for C3 plants
    Boundary layer conductance (g_b): conductance through the thin layer of still air around the leaf. It can be parameterized per unit leaf area (one-sided) as a function of leaf size (l, m) and wind speed ($u, \mathrm{m\ s^{-1}}$)

    $$g_b = 0.423(u/l)^{0.5}$$

    g_b increases with wind speed (erodes the boundary layer) and decreases with leaf size (larger leaves have a thicker boundary layer).
    Sensible heat flux occurs on both side of the leaf; the two resistances (r_b) act in parallel.
    Stomata are typically located on the lower leaf surface (hypostomatous). Some stomata are on both sides of the leaf (amphistomatous), and their resistances (r_s) act in parallel.
    For H_2O and CO_2 exchange, r_s and r_b act in series.
  - **Transpiration in land plants**
    ET is usually high in the day (highest soil-plant-atmosphere gradient in water potential), and low at night (~0 gradient before dawn) when stomata close and plant water uptake replenishes water depleted during the day.
    Predawn foliage water potential is a good indicator of soil moisture.
    As plants extract water from the soil, some critical water content is reached at which further decrease in soil water increases plant stress.

    $$-E = g_b \frac{e_a - e_s}{P} = g_s \frac{e_s - e_i}{P} = g_l \frac{e_a - e_i}{P}$$

    g_b: boundary-layer conductance for water vapor; g_s: stomatal conductance for water vapor ($\mathrm{mol\ H_2O\ m^{-2}\ s^{-1}}$); $g_l = (g_b^{-1} + g_s^{-1})^{-1}$: total leaf conductance for water vapor.
    e_a, e_s and e_i: ambient, leaf surface and intercellular water vapor pressure (Pa)
    P: surface air pressure

Generally, the intercellular space in leaves is assumed to be saturated with moisture, $e_i = e_*(T_l)$.
Leaf net photosynthesis ($\mu\mathrm{mol\ CO_2\ m^{-2}\ s^{-1}}$) can also be represented as a diffusive process:

$$A_n = g_b \frac{c_a - c_s}{1.4} = g_s \frac{c_s - c_i}{1.6} = g_l(c_a - c_i)$$

Factors 1.4 and 1.6 account for lower diffusivity of CO_2 compared with H_2O
$g_l = (1.4/g_b + 1.6/g_s)^{-1}$: total leaf conductance for CO_2
CO_2 has a slower molecular diffusion as it is heavier than water vapor. Boundary layer conductance is larger because of turbulent mixing.
Water use efficiency: the ratio of carbon gain during photosynthesis to water loss during transpiration.
Assuming $g_b \gg g_s$, then $g_l \sim g_s$, allowing cancellation of g_s:

$$\frac{A_n}{E} = \frac{c_a(1 - c_i/c_a)}{1.6D}$$

$D = (e_i - e_a)/P$: vapor pressure deficit
e: water vapor pressure
P: air pressure

- **CO_2 and stomatal conductance**
  Photosynthetic response is greater for C3 plants than C4 plants
  For C3 plants, elevated CO_2 level floods the enzyme with its preferred substrate and competitively inhibits the oxygenation reaction.
  Result: Elevated CO_2 gives C3 plants a "double benefit":
  1. More Fuel: More carbon to fix.
  2. Less Waste: Dramatically reduced photorespiration (saving energy).

  RuBisCO in C4 plants is already working in a CO_2-rich environment where photorespiration is near zero. C4 plants only get the benefit of the enhanced carbon availability.
  Franks et al. suggest that relative changes in A_n and g_s at elevated CO2 can be approximated by

$$\frac{A_n}{A_{n_0}} = \frac{(c_a - \Gamma_*)(c_{a_0} + 2\Gamma_*)}{(c_a + 2\Gamma_*)(c_{a_0} - \Gamma_*)}, \frac{g_s}{g_{s_0}} = \frac{c_{a_0}}{c_a}\frac{A_n}{A_{n_0}}, c_{a_0} = 360\ \mathrm{ppm}, \Gamma_* = 40\ \mathrm{ppm}$$

  Assumed: A_n = RuBP-limited rate; g_1 and h_s unchanged; $g_0 \ll g_s$
  Stomatal density (n_s) and stomatal pore depth (ℓ) set a theoretical maximum stomatal conductance:

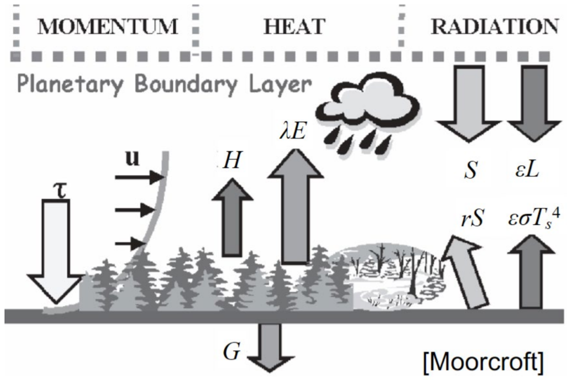
$$g_{s\ \max} = \frac{n_s a_{\max} D_w \rho_m}{\ell + \frac{\pi}{2}\left(\frac{a_{\max}}{\pi}\right)^{1/2}}$$

  Higher CO2: (1) requires fewer stomata; (2) stomatal pore can be deeper to increase water vapor diffusion length and reduce water loss.
- **Plant ecophysiology and climate**
  As ambient CO2 increases, A_n increases but g_s decreases. WUE is usually enhanced.
  As plants encounter a severe dry condition (high water vapor deficit), g_s is reduced substantially to save water, and A_n is also reduced.
  Reduced g_s decreases transpiration, reduced transpiration may dry the atmosphere (potential positive feedback: dry atmosphere $\rightarrow$ stomata close to reduce transpiration $\rightarrow$ drier atmosphere), warm the surface and thicken the boundary layer (stronger thermals grow the mixed layer).
- **Leaf area index and light penetration**
  Leaf area index (LAI, $\mathrm{m^2\ m^{-2}}$) is leaf area (one-sided) per unit area of ground. It is typically 4 to 6 $\mathrm{m^2\ m^{-2}}$ for a productive forest.
  As incident radiation penetrates through a canopy, it is attenuated by the leaves via absorption or scattering, and thus irradiance at height (z) below the canopy top is a function of cumulative LAI.
- **Canopy photosynthesis**
  GPP production efficiency model: $\mathrm{GPP} = \epsilon F f_1(T) f_2(\theta) f_3(D)$
  F = absorbed PAR, f = empirical functions scaled from 0 to 1,
  T = temperature, θ = soil moisture, D = vapor pressure deficit
  Multilayer canopy model: by explicitly calculating light penetration, photosynthesis and conductance at each canopy layer, and integrating them over all layers.
- **Big-leaf approach**: apply proper scaling (to account for light attenuation, leaf nitrogen content variations, etc.) to calculate canopy-average parameters from leaf-level parameters.
  $\mathrm{GPP} = A_l L$, A_l = leaf-level photosynthesis rate averaged over canopy ($\mu\mathrm{mol\ CO_2\ m^{-2}\ s^{-1}}$, where $\mathrm{m^{-2}}$ refers to a unit leaf area), L = leaf area index (LAI, $\mathrm{m^2\ m^{-2}}$)

$$g_{ch} = 2 g_b L\ (\text{heat}), \qquad g_c = \frac{L}{g_b^{-1} + g_s^{-1}}(\text{water vapor})$$

Topic 3 - Biogeophysical Interactions and Surface Energy Balance

- **Biogeophysical processes** involve the physical exchange of momentum, energy (heat and radiation), and water between the land surface (soil, vegetation, water) and the atmosphere.



- **Global energy balance**
  $\text{Incoming solar radiation} + \text{Atmospheric longwave radiation} = \text{Surface thermal emission} + \text{Sensible heat} + \text{Latent heat}$
- **Momentum exchange**
  Roughness of the surface (e.g., from vegetation) imposes a frictional drag on the atmosphere, which in turn imposes a stress (τ) on the surface, shaping wind velocities (u) in the boundary layer and creating vertical windshear (gradient). Wind shear and surface heating lead to dynamic and static instability, respectively, which generates turbulence, giving rise to aerodynamic conductance for land-atmosphere exchange.
  Turbulent mixing, which determines aerodynamic conductance, is an important driver of land-atmosphere exchange of heat, water and chemical substances, shaping climate and atmospheric composition at large.
- **Surface energy balance**
  $(1-r)S + \varepsilon L = \varepsilon\sigma T_s^4 + H + \lambda E + G$
  <u>Incoming radiation</u>
  S: incoming solar radiation (shortwave).
  r (albedo): the fraction of solar radiation reflected by the surface.
  L: incoming longwave radiation (infrared) from the atmosphere.
  ε (emissivity/absorptivity): the efficiency with which a surface emits or absorbs longwave radiation.
  <u>Outgoing energy</u>
  $\epsilon\sigma T_s^4$: outgoing longwave radiation emitted by the surface.
  H: sensible heat flux, the transfer of heat from the surface to atmosphere via conduction and convection driven by difference of temperature between the surface and the air.
  λE: latent heat flux, the transfer of energy from surface to atmosphere via evaporation and transpiration.
  G: ground heat flux, the transfer of heat into the soil via conduction.
  Net radiation: $R_n = H + \lambda E + G$
- **Vegetation and hydrology**
  Runoff: streamflow and groundwater outflow.
  Water balance: $\text{Runoff} = \text{Precipitaion} - \text{Evapotranspiration}.$
  Vegetation tends to retain water and reduces runoff because (1) plant tissues store water; (2) roots and organisms keep soil porous, storing more water; (3) Litter and humus retard surface outflow; (4) Suction by roots keeps soil water near surface. As a result, vegetation generally increases evapotranspiration and latent heat flux.
  Tropical rainforest → lots of soil moisture to support E → λE increases more steeply with R_n than H with R_n
  Grassland → not as much soil moisture as tropical rainforest → increase in λE with Rn is less steep than in a tropical rainforest.
- **Bowen ratio**: the ratio of sensible heat flux to latent heat flux.
  $$\beta = H/\lambda E$$
  Annual ET is ~60% of R_n, and ~65% of annual precipitation.
  Typically, tropical rainforests have a Bowen ratio of 0.1-0.3, temperate forests and grasslands have a Bowen ratio of 0.4-0.8, semiarid regions and deserts have a Bowen ratio of 2-6 and 10, respectively.
- **Albedo of land surface**: vegetation is generally less reflective of solar shortwave radiation (i.e., lower albedo) than bare lands, thus a warming effect on climate. Soil albedo decreases with increasing soil moisture, because radiation is trapped by internal reflection. Albedo varies diurnally (due to solar zenith angle) and seasonally (due to number of leaves, amount of snow cover and soil moisture).

- **Sensible heat flux**

$$H = -c_p(T_a - T_s)g_a$$

c_p: molar specific heat of moist $\approx$ $29.2 \text{ J mol}^{-1} \text{ K}^{-1}$
T_a: air temperature at reference height (K)
T_s: surface temperature (K)
g_a: aerodynamic conductance for heat and water vapor ($\text{mol m}^{-2} \text{ s}^{-1}$), typically ranging from 0.4 to 4, depending on surface roughness and meteorological conditions.

- **Latent heat flux**

$$\lambda E = -\frac{c_p}{\gamma}[e_a - e_*(T_s)]g$$

λ: latent heat of vaporization for water $\approx$ $4.444 \times 10^4\ J\ mol^{-1}$ at 15°C
c_p: molar specific heat of moist air $\approx 29.2 \text{ J mol}^{-1} \text{ K}^{-1}$
γ: psychrometric constant = $c_p P/\lambda \approx 66.5 \text{ Pa K}^{-1}$
e_a: water vapor pressure in air at reference height (Pa)
$e_*(T_s)$: saturation vapor pressure evaluated at T_s (Pa)
g: total conductance for water vapor ($\text{mol H}_2\text{O m}^{-2} \text{ s}^{-1}$)
Total conductance for water vapor: combination of aerodynamic conductance (g_a) and surface or canopy conductance (g_c) (resistances added in series):

$$1/g = 1/g_a + 1/g_c$$

g_c depends on surface dryness, soil moisture and plant physiology (e.g., stomatal conductance g_s)
Evapotranspiration rate and latent heat flux thus increase with:

- Increasing available energy (e.g., from solar radiation) and higher temperature $\rightarrow$ higher $e_*(T_s)$ $\rightarrow$ steeper gradient
- Lower humidity in air $\rightarrow$ lower e_a $\rightarrow$ steeper gradient
- Higher wind speed and turbulence $\rightarrow$ lower e_a and higheraerodynamic conductance g_a
- Wetter soil and more vegetation $\rightarrow$ higher surface conductance g_c

- **Soil heat flux**

$$G = \frac{\kappa}{\Delta z}(T_s - T_g)$$

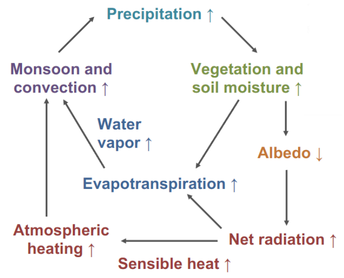
κ: thermal conductivity $W\ m^{-1}\ K^{-1}$

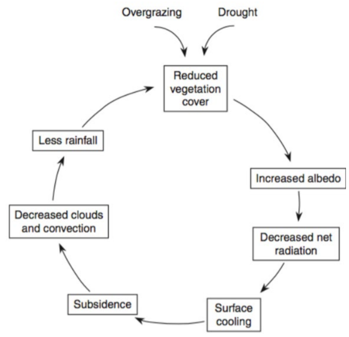
T_g: soil temperature at depth Δz

- **Biogeophysical feedback**

T $\uparrow$ $\rightarrow$ photosynthesis $\uparrow$ $\rightarrow$ albedo $\downarrow$ $\rightarrow$ T $\uparrow$ (+)
T $\uparrow$ $\rightarrow$ plant damage $\rightarrow$ photosynthesis $\downarrow$ $\rightarrow$ albedo $\uparrow$ $\rightarrow$ T $\downarrow$ (−)
T $\uparrow$ $\rightarrow$ photosynthesis $\uparrow$ $\rightarrow$ ET $\uparrow$ $\rightarrow$ T $\downarrow$ (−)
T $\uparrow$ $\rightarrow$ plant damage $\rightarrow$ photosynthesis $\downarrow$ $\rightarrow$ ET $\downarrow$ $\rightarrow$ T $\uparrow$ (+)
Radiative warming: vegetation decreases albedo (r).
Radiative cooling: vegetation takes up CO_2 by photosynthesis (biogeochemical effect, important on global scale only).
Evaporative cooling: vegetation retains water and increases roughness, thus increasing total conductance (g) and evapotranspiration (E).

- **Case studies**
  - **Hypothetical daisy world**
    A simplified model demonstrating climate-vegetation feedback. A planet with black surface and white daisies. Daisies grow best at an optimal temperature. Daisies increase planetary albedo, cooling the planet.
    The presence of life (daisies) can regulate the planet's temperature over a wide range of solar forcing.
    1. At low solar luminosity, it's too cold for daisies.
    2. At intermediate luminosity, daisies grow and cool the planet towards their optimum, creating a stable equilibrium (negative feedback loop).
    3. At high luminosity, daisies would overheat, die off, and the planet becomes even hotter, creating an unstable equilibrium (positive feedback loop).





- **Green Sahara**
  Orbital variations increased summer solar radiation in North Africa, strengthening the monsoon.
  Climate models show that the orbital forcing alone is not enough to create the observed green state. Strong positive climate-vegetation feedback was needed: More vegetation → lower albedo → more surface heating → stronger monsoon → more rain → more vegetation.
  When solar forcing gradually declined, the system did not change gradually. Instead, it underwent an abrupt transition from a wet, green state to a dry, desert state around 5,000-6,000 years ago.
  Reduced vegetation due to overgrazing and drought may activate positive biogeophysical climate feedback that further intensifies desertification. Lack of vegetation in the Sahara and Sahel sustains and reinforces their aridity.
  This suggests the existence of two stable equilibria (bistability) for the region.
- **Boreal forests**
  Strong warming effect due to their low albedo (especially masking snow).
  Their transpirational cooling is limited by needleleaf physiology and often cold soils/water availability.
  Expansion of boreal forests into tundra under climate change creates a positive feedback, amplifying Arctic warming.
- **Reforestation**
  Radiative warming and limitation on evaporative cooling due to soil water availability may partly or completely offset the effect of $CO2$ uptake.
- **Deforestation**
  Tropics: deforestation leads to strong warming because it removes a large source of evaporative cooling (high ET) and releases CO_2 (biogeochemical warming). Transpiration usually not constrained by water availability. The biogeochemical effect often dominates globally.
  Boreal/Temperate regions: historical deforestation has led to a slight cooling effect in some northern hemisphere regions. This is because the increase in albedo (from dark forest to lighter cropland or snow-covered fields) can outweigh the loss of evaporative cooling, especially in winter and spring.
  Global net effect: the biogeochemical warming from CO_2 release due to deforestation over the 20th century has likely outweighed the regional biogeophysical cooling.
- **Urban greening**
  On a local scale, planting trees can provide evaporative cooling, mitigating the urban heat island effect.
  Broadleaf trees are often better for this as they absorb less sunlight than some needleleaf trees.
  A key caveat is to avoid planting trees that emit high levels of VOCs, which can react with vehicle NOx to form ground-level ozone pollution.

Topic 4 - Biogeochemistry and Terrestrial Ecosystem Ecology

- **Biogeochemical cycles** refer to the global pathways and circulation of chemical elements (mainly C, N, P, S) through the biotic (living) and abiotic (non-living) compartments of the Earth.
- **Reservoir**: a compartment where an element is stored (e.g., atmosphere, ocean, biomass).
- **Residence time** (τ): the average time an atom or molecule spends in a given reservoir. At steady state, the total inflow into a reservoir equals the total outflow, and the residence time is given by

$$\tau = \frac{M}{F_{\text{inflow}}} = \frac{M}{F_{\text{outflow}}}$$

- **The Geological "Slow" Carbon Cycle**
  - Timescale: operates over millions to billions of years (10^6 - 10^9 yrs).
  - Processes: recycle carbon between the atmosphere, ocean, and lithosphere via geological processes. Closely linked with the hydrological cycle and acid-base chemistry.
    Weathering, Erosion, and Sedimentation: Atmospheric CO_2 is consumed.
    $$\mathrm{CaSiO_3 + CO_2 \rightarrow CaCO_3 + SiO_2}$$
    Subduction, Metamorphism, and Volcanism: CO_2 is released back to the atmosphere.
    $$\mathrm{CaCO_3 + SiO_2 \rightarrow CaSiO_3 + CO_2}$$
  - Self-Regulating feedback: higher atmospheric CO_2 leads to warmer temperatures, which accelerates the rate of weathering. Faster weathering brings down more CO_2, leading to a cooler climate and slower weathering. This feedback loop stabilizes the climate over geological timescales.
- **Weathering and Soil**
  - **Physical weathering**: physical disintegration of rocks by, e.g., swelling and shrinking due to cycles of freezing/thawing, wetting/drying, warming/cooling; root growth of plants.
  - **Chemical weathering**: minerals dissolved or chemically transformed into new minerals. Ultimately yields clay particles (silicates, iron oxides, aluminum oxides), and soil solution with dissolved ions (e.g., K^+, Na^+, Ca^{2+}, Mg^{2+}, SO_4^{2-})
    Dominant process: dissolution by carbonic acid (CO_2 dissolved in water).
    $$\mathrm{CaCO_3 + CO_2 + H_2O \rightarrow Ca^{2+} + 2HCO_3^-}\ \text{(weathering of limestone)}$$
    Other processes:
    $$\mathrm{FeOOH + 3H_2O \rightarrow Fe^{3+} + 2H_2O + 3OH^-}\ \text{(hydrolysis)}$$
    $$\mathrm{CaSO_4 + 2H_2O \rightarrow CaSO_4 \bullet 2H_2O}\ \text{(hydration; product more soluble)}$$
    $$\mathrm{Fe^{2+} \rightarrow Fe^{3+} + e^-}\ \text{(oxidation and reduction)}$$
  - Plants require nutrients like nitrate (NO_3^-) and phosphate (PO_4^{3-}) for assimilation into organic matter. Availability of nutrients to plants often depends on soil conditions such as pH and cation exchange capacity.
    Nitrogen Assimilation: $\mathrm{NO_3^- + H^+ + H_2O \rightarrow NH_{3\,(OM)} + 2O_2}$ (Nitrogen is reduced)
    Phosphorous Assimilation: $\mathrm{H_2PO_4^- + H^+ \rightarrow H_3PO_{4\,(OM)}}$
  - **Sources of Nutrients**
    Nitrogen (N): the ultimate source is the atmosphere, fixed by biological or abiotic processes. For proteins, chlorophyll and enzymes.
    Other nutrients (e.g., P, K^+, Ca^{2+}): Derived from the chemical weathering of rocks and minerals.
    Clay and organic matter particles have negative charges that bind cations.
    Cation Exchange Capacity (CEC): the ability of soil to hold and exchange positively charged ions (cations), higher at higher pH. High CEC improves potential fertility by preventing nutrients from being washed away.
    - Acid rain can mobilize nutrients (like Ca^{2+}) by replacing them with H^+ on soil particles, leading to their loss via leaching.
    - Plants have a strong preference for NO_3^- to NH_4^+, even at the expense of extra energy to reduce NO_3^- to NH_3(OM). Reason: high concentration of ammonium can disrupt the process of photosynthesis in chloroplasts, acidify cells by assimilating into amino acids, and compete with other essential cations for uptake. Plant can also flexibly transport nitrate to the leaves during daytime as there is abundant access to energy, the surplus of reducing power (ATP and NADPH) can be used to convert NO_3^- to NH_3 at a very low marginal cost.
  - Cation Affinity Order: $Al^{3+} > H^+ > Ca^{2+} > Mg^{2+} > K^+ = NH_4^+ > Na^+$
  - **Translocation**: recycling of nitrogen and phosphorous required for plant growth via the withdrawal of nutrients from senescing leaves and subsequent storage within the plant.
- **Processes in Contemporary Carbon Cycle**
  Photosynthesis: $\mathrm{CO_2 + H_2O \rightarrow CH_2O + O_2}$
  Respiration / Decomposition / Combustion: $\mathrm{CH_2O + O_2 \rightarrow CO_2 + H_2O}$
  Fossil Fuel Combustion:
  Coal: $\mathrm{C + O_2 \rightarrow CO_2}$
  Petroleum: $\mathrm{CH_2 + 1.5O_2 \rightarrow CO_2 + H_2O}$
  Natural gas: $\mathrm{CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O}$
  Ocean Uptake: $\mathrm{CO_2(g) + CO_3^{2-} + H_2O \rightarrow 2HCO_3^-}$
  Calcite Formation: $\mathrm{Ca^{2+} + 2HCO_3^- \rightarrow CaCO_3(s) + CO_2(aq) + H_2O}$

- **Biological Primary Production**

$$\mathrm{a\ CO_2 + b(H^+ + NO_3^-) + c(H^+ + H_2PO_4{}^-) + (a+b)H_2O \rightarrow (CH_2O)_a(NH_3)_b(H_3PO_4)_c + (a+2b)O_2}$$

  - **Marine C:N:P (Redfield Ratio)**: Relatively constant at 106:16:1.
  - **Terrestrial plants C:N:P**: average is 882:9:1. The higher C content reflects more structural cellulose.
- **Allocation**: the partitioning of NPP to different plant parts (leaves, stems, roots, reproduction) to balance conflicting needs for resources. Plants allocate more to the part acquiring the most limiting resource.
  - **Leaves**: to capture light and absorb CO_2 for new growth
  - **Stems**: for structural support, water transport, and competition for light
  - **Roots**: for soil water and nutrient uptake, and food storage
  - **Flowers, fruits, seeds**: for reproduction and dispersal

Stress	Root growth	Foliage growth
Shade	Reduced	Increased
Drought	Increased	Reduced
Cold temperature	Increased	Reduced
Nitrogen deficiency	Increased	Reduced
Nitrogen surplus	Reduced	Increased

- **Phenology**: the study of seasonal plant life cycles (onset and duration at different phases of plant development, e.g., leaf emergence, senescence). Controlled by temperature, precipitation, soil moisture, and photoperiod.
- **Decomposition, Mineralization, and Immobilization**
  - **Litterfall**: Transfers carbon and nutrients from living biomass to the soil.
  - **Litter**: easily recognizable, undecomposed materials (e.g., leaves, twigs, branches, carcasses).
  - **Humus**: layer underneath formed by highly decomposed organic materials.
  - Initially, microbial demand for nutrients may exceed that can be supplied by the decaying materials. There can be a net increase in the nitrogen amount in the decomposing litter as microbes take up nutrients from the soil and incorporate into litter. As more materials decompose, microbial demand diminishes and there is a net mineralization of nutrients available for plant growth.

    Mineralization: The microbial process of converting organic forms of nutrients (e.g., N, P) back into inorganic forms (e.g., NH_4^+, NO_3^-, PO_4^{3-}) that plants can use.

    Immobilization: The temporary uptake of inorganic nutrients by microbes from the soil solution during decomposition. This reduces nutrient availability for plants.
  - **Decomposition Formula**:

$$\mathrm{(CH_2O)_a(NH_3)_b(H_3PO_4)_c + (a+2b)O_2 \rightarrow\ a\ CO_2 + b(H^+ + NO_3^-) + c(H^+ + H_2PO_4{}^-) + (a+b)H_2O}$$

  - **Decay Rate (k)**: The rate at which organic matter decomposes. Depends on temperature, moisture, pH, and litter quality (e.g., C:N ratio, lignin content).

$$M = M_0 e^{-kt}$$

    Cold climates have low k (~0.1), warm humid climates have high k (~1-4).
  - **Net Mineralization**: The net release of inorganic nutrients available for plants.

$$\text{Net Mineralization} = \text{Gross Mineralization} - \text{Immobilization}$$

$$\text{Gross Mineralization} = (1 - e^{-k\Delta t})/\mathrm{CN_{litter}}$$

$$\text{Immobilization} = g(1 - e^{-k\Delta t})/\mathrm{CN_{Soil\ Organic\ Matter}}$$

  - **Litter Quality**: affects the rate of microbial decomposition

    High C:N ratio (low quality, e.g., woody debris): Decomposes slowly. Microbes may need to take in N from the soil, leading to immobilization.

    Low C:N ratio (high quality, e.g., many deciduous leaves): Decomposes quickly. More N is released than the microbes need, leading to net mineralization.
- **Modeling Ecosystem Carbon and Nitrogen Cycling**

  Each carbon pool j has a pool size x_i and a baseline turnover time τ_j, related to the decay rate k_j by $\tau_j = 1/k_j$.

  Flux F_{ij} is from pool j to pool i, specified by $F_{ij} = x_j(1 - e^{-k_j\Delta t}) \times f_{ij} \times (1 - r_{ij})$.

  Each carbon pool has an associated nitrogen pool $\mathrm{C{:}\,N}$ ratio, depending on the soil nitrogen content.
- **Dynamics in Forest Ecosystems**
  - **Forest dynamics** are driven by individual level processes:

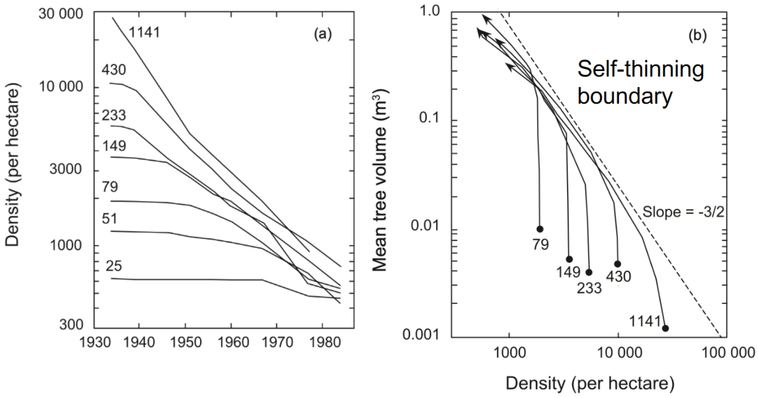
    <u>Reproduction</u>: seed production, dispersal, seed bank. Most seeds never germinate.

    <u>Germination & establishment</u>: only a tiny fraction survives seedling stage (high mortality from competition, herbivory, shade).

    <u>Growth</u>: interference competition for light, water, nutrients. Larger trees suppress smaller ones.

    <u>Mortality</u>: (i) stand-replacing (fire, wind, insects); (ii) gap phase (canopy gap created by single tree fall cause canopy gap release of understory); (iii) age related (senescence, hydraulic failure)
  - **Succession**:

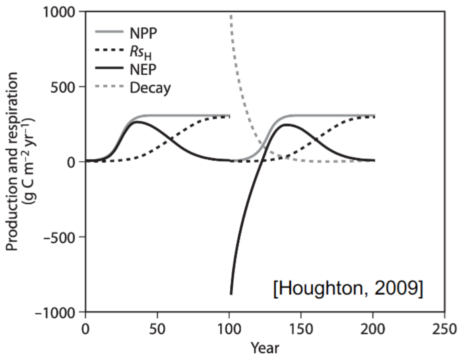
    Primary succession: Colonization of new land not previously vegetated (following volcanic eruption, glacial melting, floodplain and sand dune formation, etc.).
    - At early stage of stand development, plants do not interfere with growth of neighbors, mean tree mass and total biomass can increase without a decline in density (toward the self-thinning boundary line).
    - Once the canopy closes and competition starts, self-thinning occurs, i.e., further increases in biomass can only be achieved via decreases in density. This is usually because as larger plants use up majority of resources, smaller plants are suppressed and may die, and density progressively declines.

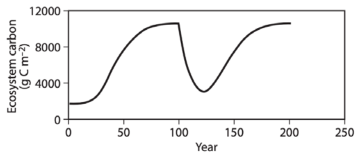


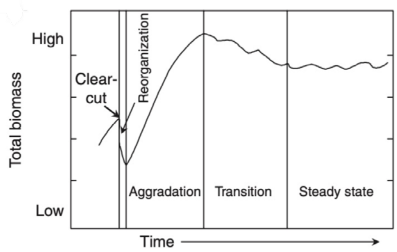
For a self-thinning population with density N:
Mean plant mass $\overline{m} = cN^{-3/2}$
Total biomass: $M = cN^{-1/2}$

- As a forest matures, productivity generally plateaus due to canopy light saturation at high LAI, resource limitation and competition. Productivity then typically declines, likely due to increased respiration from woody biomass, decreased photosynthetic capacity, aging, etc., before finally reaching steady state.

Secondary succession: Regrowth of vegetation on previously vegetated land following a disturbance (e.g., fire, windthrow, logging, clear-cutting, farm abandonment).
After a stand-replacing disturbance, a forest goes through distinct carbon phases.
<u>Reorganization</u> (year 0-10): NEP is negative, rate of C loss from decay of accumulated biomass is greater than the rate of C gain through NPP. General decline in annual mean surface albedo as forest recovers. Dead wood decomposes; vegetation regrows slowly; net carbon loss. Bowen ratio (H/λE) is initially high, then decreases as forest recovers, but finally increases again as forest ages.
<u>Aggradation</u> (year 10-80): NEP is positive, rapid biomass accumulation; canopy closes; LAI increases.
<u>Transition</u> (year 80-200): NEP is near zero, slow growth; respiration from woody biomass increases; gap-phase dynamics.
<u>Steady state</u> (year 200+): NEP is zero, old-growth forest, carbon gain is roughly equal to the loss; small disturbances maintain balance.







Early v.s. late successional species

	Early succession	Late succession
Seeds		
Number	Many	Few
Size	Small	Large
Dispersal	Wind, birds	Gravity, mammals
Dormancy	Long	Short
Germination	Light enhanced	Not light enhanced
Photosynthesis		
Light saturation intensity	High	Low
Light compensation point	High	Low
Efficiency at low light	Low	High
Maximum rate	High	Low
Stomatal conductance	High	Low
Morphology		
Root-to-shoot ratio	Low	High
Size at maturity	Small	Large
Structural strength	Low	High
Lifespan	Short	Long

- Global annual GPP is estimated to be ~120 Pg C yr^{-1}. Roughly half of GPP goes to autotrophic respiration, and the other half is returned to the environment through heterotrophic respiration, leaching and disturbances.
- Despite slow litter production, very slow decomposition in boreal forests due to cold climate enhances their soil carbon storage. The thick forest floor further hinders heating in summer.
- In broadleaf forests, warm temperature enhances mineralization and nitrogen availability, with a faster litter quality, NPP, and turnover rate.

- **Marine Biogeochemistry**

Dissolution of atmospheric CO2 in seawater: $\mathrm{CO_2 + CO_3^{2-} + H_2O \rightleftharpoons 2HCO_3^-}$, $K' = \frac{[\mathrm{HCO_3^-}]}{p_{\mathrm{CO_2}}[\mathrm{CO_3^{2-}}]}$

Photosynthesis: $\mathrm{CO_2(aq) + H_2O \rightarrow CH_2O_{(OM)} + O_2}$

Nitrate uptake: $\mathrm{NO_3^- + H^+ + H_2O \rightarrow NH_{3(OM)} + 2O_2}$

Phosphate uptake: $\mathrm{HPO_4^{2-} + 2H^+ \rightarrow H_3PO_{4(OM)}}$

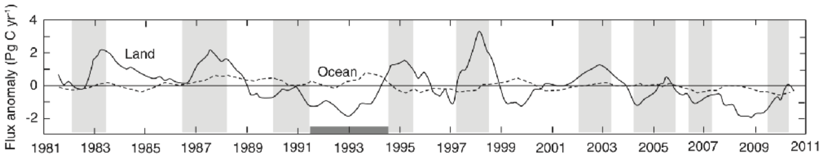
Calcite formation: $\mathrm{Ca^{2+} + 2HCO_3^- \rightarrow CaCO_3(s) + CO_2(aq) + H_2O}$

Cyanobacteria are the oldest known bacteria to produce oxygen through photosynthesis;
Prochlorococcus are the most abundant cyanobacteria;
Coccolithophores are single-cell shell-making algae.

- The ability of ocean to absorb $\mathrm{CO_2}$ is limited by the availability of $\mathrm{CO_3^{2-}}$, for calcifying organisms to build their shells and skeletons.
- Warming of seawater (shifting phase equilibrium to gas phase) and depletion of $\mathrm{CO_3^{2-}}$ at higher $\mathrm{p_{CO_2}}$ reduce the ocean's ability to dissolve more $\mathrm{CO_2}$ in future.
- Air-sea exchange of $\mathrm{CO_2}$ is determined by the difference between equilibrium $p_{\mathrm{CO_2}}$ at the sea surface and $p_{\mathrm{CO_2}}$ of the atmosphere.
- $[\mathrm{DIC}] = [\mathrm{CO_2}] + [\mathrm{HCO_3^-}] + [\mathrm{CO_3^2}-], [\mathrm{POC}] \approx [\mathrm{CH_2O}], [\mathrm{PIC}] \approx [\mathrm{CaCO_3}]$
- **Solubility pump**: Dissolved organic carbon (DOC) and inorganic carbonate species are "pumped" down the deep ocean by the downwelling of cold, dense water in polar regions.
- **Biological pump**: Dead bodies and fecal pellets sink as particulate organic carbon (POC) together with $\mathrm{CaCO_3}$ shells to the deep ocean, where they are remineralized to $\mathrm{CO_2}$ or become sediments. Iron can be a major limiting nutrient in the open ocean, affecting the biological pump.

Topic 5 – Global Carbon Cycle and Biogeochemical Climate Feedbacks

- **Glacial-Interglacial Cycles**
  During glacial periods: Ocean held more CO_2 (colder water, circulation changes); terrestrial biosphere held less (cooler, drier climate).
  Interglacial periods: CO_2 relatively stable until the industrial era.
- **Observed Atmospheric CO_2 Concentration**
  Keeling curve: increases by ~1.5–2 ppm per year in recent decades.
  Theoretically, 2.13 Pg of carbon emission into the atmosphere is equivalent to increasing CO2 level by 1 ppm.
  Over the industrial era, human activities emitted 555 $\pm$ 85 PgC from fossil fuel combustion, but measured CO2 only rose by ~110 ppm, as only roughly half of CO_2 emitted remained in the atmosphere. The other half was taken up together by the oceans and the terrestrial biosphere-soil system.
- **Seasonality of Carbon Fluxes**
  Mauna Loa and South Pole records show seasonal cycling pattern, where photosynthesis as a sink for CO_2 from spring to summer (growing season), but net respiration from fall to winter (especially for deciduous forests at northern middle-high latitudes). Seasonal cycle is most pronounced in northern deciduous forests, less in southern hemisphere due to less land area and deciduous forests.
- **Interannual Variability of Carbon Fluxes**

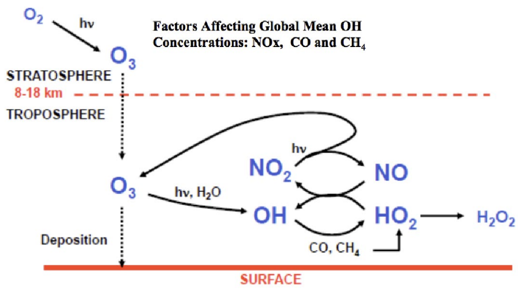
  

  El Niño: land-to-atmosphere carbon fluxes are generally more positive. Reason: drought and warming in tropical regions reduce soil moisture, which then limit photosynthesis and lower NPP. Higher temperatures and drier conditions accelerate heterotrophic respiration and decomposition of soil organic matter. Increased frequency and intensity of forest fires (e.g., in Indonesia and Amazon) release large amounts of stored carbon directly into the atmosphere.
  Mt. Pinatubo eruption: injection of tons of SO_2 into the stratosphere formed sulfate aerosols that persisted for ~2-3 years. These aerosols scattered incoming solar radiation and reduced surface temperatures. Diffuse radiation penetrates deeper into plant canopies and increases photosynthesis, whereas lower temperatures can reduce respiration and allow more NPP to be stored as biomass and soil organic matter.
- **Trends in Global Carbon Cycle**
  In the last decade, human activities emit ~11 Pg C yr^{-1}, atmospheric CO_2 increase by ~5 Pg C yr^{-1}, ocean uptake is ~3 Pg C yr^{-1}, and land biosphere uptake is ~3 Pg C yr^{-1}.
  Land use source is mainly due to deforestation in the tropics.
  Both the ocean and land carbon sinks generally increase over the past 60 years, with great interannual variability.
  The ocean sink has increased due to a larger concentration gradient across the air-sea interface, increasing solubility.
  The land sink has increased primarily due to CO_2 fertilization and forest regrowth in temperate regions.
  Tropical forests are carbon neutral due to the offset by deforestation in mature Amazonian and African tropical forests.
  Regrowth of temperate forests in North America and Europe represents the largest residual terrestrial sink.
- **CO_2 Fertilization and Nitrogen Limitation**
  Progressive nitrogen limitation: enhanced plant growth under elevated CO_2 requires more nitrogen; plants sequester nitrogen in their biomass and in soil organic matter; over time, the soil mineral N pool is depleted; this reduces the initial CO_2-induced NPP increase.
  Long-term increase in lank sink may be limited due to: (i) forest maturation (old, mature forests have lower NPP than young, fast-growing forests); (ii) ecosystem respiration catches up (autotrophic and heterotrophic increases with biomass and eventually balancing NPP); (iii) progressive nitrogen limitation (soil N depletion reduces CO_2 fertilization response); (iv) allocation (if extra NPP is allocated to leaves, the net carbon sink is less effective).
- **Effects of Warming on Soil and Vegetation**
  Decomposition $\uparrow \rightarrow$ soil C loss; N mineralization $\uparrow \rightarrow$ vegetation C gain (N may be lost via leaching or denitrification).
  Longer growing season + higher T $\rightarrow$ NPP $\uparrow$ (enough water), T $\uparrow \rightarrow$ water stress $\rightarrow$ NPP $\downarrow$ (limited water).
  Indirect effects: wildfires, insect outbreaks, extreme events increase in frequency, permafrost thaw releases carbon.
  Recent net midlatitude terrestrial carbon sink is likely temporary and will diminish in importance as a new steady state is approached.
  Soil and organic carbon, thawing of permafrost, ecosystem disturbances will become important CO_2 sources in the future.
- **Wildfires and Climate**
  Climate anomalies during El Niño events increase fire occurrence and carbon emissions in tropical forests worldwide.
  Fire occurrence will likely increase under climate change with warmer, drier environment, and with more frequent extremes. Chemicals (NOx, CO, VOCs, primary particulate matter) released worsen air quality (O_3, PM). Black carbon released can be transported and deposited to remote glaciers and sea ice, absorbing more solar radiation and worsening their melting.

- **Insect Outbreaks and Forest Carbon Cycle**
  Cyclical insect outbreaks are important natural disturbances to forests. It reduces growth, kills trees, and transfers much living biomass to dead organic matter, making it decompose faster and more susceptible to fires.
- **Nitrogen Decomposition and Fertilization**
  Present day reactive nitrogen (Nr) deposition onto land is ~63 Tg N yr^{-1}.
  Major sources: fossil fuel combustion (NOx) and agriculture (NH_3 from fertilizer and livestock).
  Carbon-only models (without N cycle) overestimate the land sink because they ignore progressive nitrogen limitation.
  Including N cycling reduces the projected future sink, especially under high CO_2.
- **Ocean Acidification**
  Rising CO_2 contributes to a declining trend in ocean surface pH over the past two decades, with an accompanying decline in $[CO_3^{2-}]$. Biological pump may be less effective.
  Coral bleaching: loss of photosynthetic algal protists through either expulsion or loss of algal pigmentation.
  K' decreases as temperature increases, i.e., less CO_2 can dissolve as the surface ocean becomes warmer $\rightarrow$ positive feedback. As the ocean is acidified upon carbon uptake, CO_3^{2-} becomes more depleted, limiting the ability of ocean to further absorb CO_2 $\rightarrow$ negative feedback due to ocean uptake is reduced. Warming also makes the surface ocean more stratified, which suppresses upwelling and nutrient mixing from deep water, and hinders phytoplankton growth.
- **Ocean Biogeochemical Feedbacks**
  T $\uparrow \rightarrow$ CO_2 solubility $\downarrow \rightarrow$ $CO_2 \uparrow \rightarrow$ T $\uparrow$ $(+)$
  T $\uparrow \rightarrow$ upwelling $\downarrow \rightarrow$ phytoplankton $\downarrow \rightarrow$ $CO_2 \uparrow \rightarrow$ T $\uparrow$ $(+)$
  $CO_2 \uparrow \rightarrow$ pH & $[CO_3^{2-}]$ $\downarrow$ ($\rightarrow$ phytoplankton $\downarrow$) $\rightarrow$ $CO_2 \uparrow$ $(+)$
  T $\uparrow \rightarrow$ desertification $\uparrow \rightarrow$ dust $\uparrow \rightarrow$ iron deposition $\uparrow \rightarrow$ phytoplankton$\uparrow \rightarrow$ $CO_2 \downarrow \rightarrow$ T $\downarrow$ $(-)$
  T $\uparrow \rightarrow$ desertification $\uparrow \rightarrow$ dust $\uparrow \rightarrow$ iron deposition $\uparrow \rightarrow$ phytoplankton$\uparrow \rightarrow$ dimethyl sulfide $\uparrow \rightarrow$ sulfate $\uparrow \rightarrow$ albedo $\uparrow \rightarrow$ T $\downarrow$ $(-)$
- **Carbon-Induced Climate Feedbacks**
  $$\Delta C_L = \beta_L \Delta C_A + \gamma_L \Delta T, \Delta C_O = \beta_O \Delta C_A + \gamma_O \Delta T$$
  Land and ocean carbon stocks are driven by two types of feedback: one by CO_2 concentration, the other by CO_2-induced climatic changes. Generally $\beta_{L/O} > 0$ (negative feedback), $\gamma_{L/O} < 0$ (positive feedback).

Topic 6 – Nitrogen Cycle, Atmospheric Chemistry and Climate

- **Internal Nitrogen Cycling in Soils**
  Nitrate Assimilation: $NO_3^- + H^+ + H_2O \rightarrow NH_{3(OM)} + 2O_2$
  Coupled with oxygenic photosynthesis (requires energy to reduce NO_3^- to NH_3).
  Plants strongly prefer NO_3^- to NH_4^+ as NO_3^- is mobile in soil solution and can be taken up even from dilute concentrations. Moreover, NH_4^+ is often toxic at high concentrations and competes with K^+ and Ca^{2+} uptake.
  Ammonification (mineralization): $NH_{3\,(OM)} + H^+ \rightarrow NH_4^+$
  Performed by bacteria and fungi during decomposition of organic matter, releases plant-available NH_4^+.
  Some NH_4^+ volatilizes to the atmosphere: $NH_4^+ + OH^- \rightleftharpoons NH_3 + H_2O$
  Nitrification: (Nitrosomonas) $2NH_4^+ + 3O_2 \rightarrow 2NO_2^- + 2H_2O + 4H^+$; (Nitrobacter) $2NO_2^- + O_2 \rightarrow 2NO_3^-$
  This process is done by chemoautotrophic bacteria.
  H^+ is produced and acidifies soils. Small amounts of NO and N_2O are produced as byproducts.
- **External Nitrogen Cycling with the Atmosphere**
  Denitrification: $4NO_3^- + 4H^+ + 5CH_2O \rightarrow 2N_2 + 7H_2O + 5CO_2$
  Anaerobic respiration by bacteria (e.g., Pseudomonas) in anoxic conditions (waterlogged soils, sediments).
  Converts NO_3^- back to N_2 gas – the major natural pathway for removing reactive N from ecosystems.
  Byproducts: NO and N_2O (especially under incomplete denitrification).
  Biological Nitrogen Fixation (BNF): $N_2 + 2CH_2O + 2H_2O + 2H^+ \rightarrow 2NH_4^+ + H_2 + 2CO_2$
  By marine cyanobacteria and symbiotic bacteria in legume plants.
  Requires much energy directly from photosynthesis or from respiration of organic matter in local anoxic conditions.
  Lightning: $N_2 + O_2 \rightarrow 2NO,\ 2NO + O_2 \rightarrow NO_2,\ NO_2 + OH \rightarrow HNO_3 \rightarrow HNO_3(aq)$
  Natural source of reactive nitrogen, especially in tropical thunderstorms.
  Soil moisture is the primary determinant of the relative loss rates of NO, N_2O and N_2.
- **Human Effects on the Nitrogen Cycle**
  Fossil Fuel Combustion: $N_2 + O_2 \rightarrow 2NO,\ 2NO + O_2 \rightarrow NO_2,\ NO_2 + OH \rightarrow HNO_3$
  Production of NOx can lead to tropospheric ozone formation, acid rain and particulate matter.
  Haber-Bosch Process (Industrial N_2 Fixation): $N_2 + 3H_2 \rightarrow 2NH_3$, primarily used to make NH_4^+ fertilizers.
  Cultivation-Induced Biological Nitrogen Fixation (BNF): from plant legumes and fertilizers.
  Excess NH_4^+ acidifies soil, worsens leaching and leads to more soil emission of NH_3, NO and N_2O.
- **Global Atmospheric Nitrogen Budget**
  Total N deposition from global Nr emissions is ~105 Tg N yr^{-1}. Humans have doubled the global rate of reactive nitrogen creation.
  Hotspots: eastern US, Europe, India, eastern China.
  Human beings are now responsible for roughly half of all nitrogen fixation.
- **Mitigating Human Effects on Nitrogen Cycle**
  Improve combustion efficiency (reduce NOx from vehicles and power plants by installing catalytic converters).
  Use fertiliser more efficiently (right source, rate, time, pace, precision agriculture, slow-release formulations).
  Reduce meat consumption and optimize feed management (meat production is N-intensive, lower crude protein).
  Capture and recycle manure (reduce NH_3 volatilization).
  Waste water treatment with biological nutrient removal allowing for potential recycling.
  Wetlands / denitrification barriers to remove NO_3^- from runoff.
- **Tropospheric Ozone Production**



Initiation: $O_3 + h\nu \rightarrow O + O_2, O + H_2O \rightarrow 2OH$
O_3 formation in presence of NOx by CO:

$$CO + OH \rightarrow CO_2 + H$$
$$H + O_2 \rightarrow HO_2$$
$$HO_2 + NO \rightarrow OH + NO_2$$
$$NO_2 + h\nu \rightarrow NO + O$$
$$O + O_2 \rightarrow O_3$$

By increasing [OH], higher NOx emission leads to more CH_4 destruction which offsets warming.
Surface ozone is a major air pollutant; its formation is enhanced by high temperature and sunlight (summer smog).
Health effects: respiratory illness, reduced lung function, premature death.
Crop damage: reduces agricultural yields (wheat, soybean, rice).
Greenhouse gas: O_3 itself is a strong greenhouse gas (third most important after CO_2 and CH_4).

- **Aerosol Formation**
Main components: sulfate, nitrate, ammonium, black carbon,organic carbon, mineral dust, sea salt.
Anthropogenic sources: fossil fuel combustion, agriculture, biomass burning.
Natural sources: biogenic VOCs, dust, sea salt, volcanoes.
$$\mathrm{NO_2 + OH \rightarrow HNO_3 + NH_3 \rightleftharpoons NH_4NO_3}$$
Sulfate nitrate ammonium aerosols (NH_4NO_3, $(NH_4)_2SO_4$) are generally reflective, which has a cooling effect on climate (increases albedo).
Black carbon is highly absorptive of solar radiation, which causes a warming effect. Black carbon can also deposit onto snow and ice that greatly reduces surface albedo.
Mineral dust can be both absorptive (warming) and scattering (cooling) depending on its origin and atmospheric processing. It also provides nutrients (e.g., Fe, P) for marine and terrestrial ecosystems downwind of source regions.
Wet deposition (scavenging by rain) is a major pathway for removing aerosols.
Photochemical reactions involving H_2O and dry deposition of O_3 by plants can also act as a major sink for ozone.
Deposited NH_4NO_3 fertilizes ecosystems, which can increase plant growth (CO_2 sink) but also cause eutrophication of aquatic systems.
- **Influence of Reactive Nitrogen on Climate**
<u>Effect on CO_2</u>: Nr deposition increases land CO_2 uptake
<u>Effect on CH_4</u>: Higher NOx emissions decrease CH_4 lifetime and Nr deposition decreases soil CH_4 uptake
<u>Effect on N_2O</u>: emissions increase
<u>Effect on tropospheric ozone</u>: ozone is a GHG and reduces land CO_2 uptake
<u>Effect on aerosols</u>: increased concentration, reflects incoming solar radiation
Net effect is cooling in the past and the future but is highly uncertain.
Conifer forests at high northern latitudes are projected to be replaced by broadleaf trees due to warming and CO_2 fertilization. This causes a reduction in surface ozone of up to 10 ppb in the next century as broadleaf trees have higher stomatal conductance per unit leaf area and larger leaf area index, giving higher ozone deposition velocities.
Deforestation in low-NOx regions (e.g. Southeast Asia, Central Africa) worsen ozone air pollution; However, reduced biogenic VOC emissions (e.g. eastern North America, western Europe) reduce ozone in high-NOx regions.
- **Biogenic Volatile Organic Compounds**
Reactive in the atmosphere – influence ozone, aerosols, and the oxidative capacity (OH levels).
Known functions include: protection against heat stress (isoprene stabilizes thylakoid membranes), protection against oxidative stress (scavenge reactive oxygen species), defense against herbivores and pathogens (monoterpenes, sesquiterpenes), chemical signaling (between plants, or to attract predators of herbivores), thermotolerance – isoprene emissions increase sharply with leaf temperature.
Important BVOCs: isoprene (1/2 of all non-CH_4 VOCs, roughly same as CH_4), α-pinene, methanol, acetone.
- **Effects of Vegetation on Climate and Air Quality**

Effect	**Mechanism**	**Climate / air quality impact**
Lower albedo	Forests are darker than bare land or snow	Warming (absorbs more solar radiation)
Photosynthesis	CO_2 uptake	Cooling (reduces greenhouse effect)
Transpiration	Latent heat flux	Cooling at surface, but complex boundary-layer effects
Dry deposition	Uptake of O_3, SO_2, NOx via stomata	Improves air quality (removes pollutants)
BVOC emissions	O_3 and aerosol precursors	Worsens ozone and PM (depending on NOx)

In urban areas, vehicle emission control and planting low-VOC-emitting trees will help with the ozone pollution.
- **Radiative and Biogeochemical Effects of Aerosols**
Fires are a major source of both gases and particulate matter (aerosols) that impact the climate through multiple pathways.
Primary Emissions: Fires release greenhouse gases ($\mathrm{CO_2, CH_4, N_2O}$) and carbonaceous aerosols (Organic and Black Carbon).
Radiative Interactions:
  <u>Direct</u>: Aerosols interact with incoming solar radiation (scattering or absorption).
  <u>Indirect</u>: Aerosols interact with clouds, changing their reflectivity (albedo) and lifetime.
- **Afforestation-Induced Climate Feedbacks**
Trees sequester $\mathrm{CO_2}$, lowering atmospheric concentrations; Trees emit BVOCs. (cooling)
Increased moisture release leads to more water vapor; Darker forest absorb more sunlight than open land (warming).
- **Ozone and BVOC Feedbacks**
atmospheric $CO_2 \uparrow \rightarrow$ stomatal conductance $g_s \downarrow \rightarrow$ less O_3 is absorbed by plants $\rightarrow$ atmospheric $O_3 \uparrow \rightarrow$ higher O_3 damages, photosynthesis $\downarrow \rightarrow$ plants absorb less $CO_2 \rightarrow$ atmospheric $CO_2 \uparrow$
$T \uparrow \rightarrow$ BVOC $\uparrow \rightarrow$ OH $\downarrow \rightarrow CH_4 \uparrow \rightarrow$ warming $\uparrow \rightarrow T \uparrow$
$T \uparrow \rightarrow$ soil $NO_x \uparrow \rightarrow O_3 \uparrow \rightarrow$ plant damage $\rightarrow CO_2 \uparrow \rightarrow T \uparrow$
Increased CO_2 causes BVOC emission to increase, which can also cause more production of ozone.
- **Aerosol-Mediated Biogeochemical Feedbacks**
temperature $\uparrow \rightarrow$ BVOC $\uparrow \rightarrow$ SOA formation $\uparrow \rightarrow$ albedo $\uparrow \rightarrow$ temperature decrease $\downarrow$
$CO_2 \uparrow \rightarrow$ BVOC $\downarrow \rightarrow$ SOA $\downarrow \rightarrow$ albedo $\downarrow \rightarrow T \uparrow$
Photosynthesis $\uparrow \rightarrow$ BVOC $\uparrow \rightarrow$ SOA $\uparrow \rightarrow$ albedo $\uparrow \rightarrow T \downarrow$

Topic 7 – Ecosystem Modeling and Ecosystem Management

- **Evolution of Land Surface Models**

  First Generation Models

  No explicit vegetation representation.
  Prescribed albedo, surface roughness, no canopy radiative transfer.
  Single bulk aerodynamic conductance.
  Latent heat flux modulated by a soil wetness factor (β).
  Key equations:

$$H = -c_p(\theta_a - T_s)\, g_{ah}$$

$$\lambda E = -\frac{c_p}{\gamma}\left[e_a - e_s(T_s)\right] g_{aw} \cdot \beta$$

  Second Generation Models

  Single-layer, two-source canopy (vegetation + soil separately).
  Include biological controls on stomatal conductance (g_{sw}).
  Example: Jarvis empirical model:

$$g'_{sw} = g'_{sw}(I) \cdot f_1(T) \cdot f_2(D) \cdot f_3(\psi) \cdot f_4(c_a)$$

  where:
  I = light (photosynthetically active radiation)
  T = temperature
  D = vapour pressure deficit
  ψ = leaf water potential
  c_a = atmospheric CO_2 concentration
  Limitation: Direct coupling between photosynthesis and stomatal conductance.

  Third Generation Models

  Directly linked photosynthesis and stomatal conductance.
  Example: Farquhar-Ball-Berry model, for C3 plants.

$$g_s = m \cdot \frac{Ah_s}{c_s} + g_0,$$

  where A = assimilation rate, h_s = relative humidity at leaf surface, c_s = CO_2 at leaf surface.
  Required more plant physiological and biogeochemical representation and parameters ($V_{c\,max}$, J_{max}, etc.).
  Integration of leaf photosynthesis over canopy via detailed scaling parameterization (e.g., using $f_{sun}(x)$, $f_n(x)$).
  Some models include photosynthesis mainly for the sake of computing biogeophysical fluxes and surface energy balance.
  Current generation of Earth system models may further include dynamic biogeochemical and ecosystem processes, e.g., carbon and nitrogen cycles, vegetation dynamics, ecosystem disturbances, land use and urbanization.

- **Dynamic Global Vegetation Models (DGVMs)**

  Top-down Biogeographical Models

  - Approach: Correlate biome distribution with climate variables (temperature, precipitation).
  - Example: Holdridge life zones, Koppen–Geiger.
  - Limitation: Assumes climate–vegetation equilibrium; cannot predict future changes under novel climates.

  "Big Leaf" Model (Bottom-up approach)

  - Model fast dynamics: photosynthesis (minutes to months) for each PFT.
  - PFTs compete for light, water, nutrients.
  - Slow dynamics (months to decades): growth, mortality, recruitment.
  - Ecosystem structure (LAI, PFT cover) changes accordingly.

  Merits of DGVM approach: mechanistic (uses actual physiological responses), realistic predictions (works where future climates breaks past correlations), captures feedbacks, physically and biologically consistent.
  Challenges of DGVMs: lack of diversity, as each biome is modeled by one single PFT, parameterization is poorly constrained with coarse grid representations.

- **Land-Atmospheric Coupling in Earth System Models**

  Mosaic approach (current standard)

  - A grid cell is divided into patches (e.g., forest, grassland, crop, bare soil).
  - Each patch is modelled separately; grid-cell average = weighted average.
  - Allows heterogeneity without sub-grid climate feedbacks (except aggregated fluxes).
  - Earlier method: Single dominant vegetation type per grid cell (too crude).

  Global Land Cover Data

  - Maps: Satellite-derived land cover (e.g., MODIS, ESA CCI).
  - Used to prescribe initial PFT fractions and LAI.

  Asynchronous coupling is computationally cheap but ignores fast feedbacks and assumes equilibrium.
  Synchronous coupling can be computationally expensive while captures transient feedbacks and disturbances.

- **Geoengineering and Carbon Capture**

  Solar Radiation Management (SRM)

Goal: Reflect a small fraction of incoming solar radiation to cool the planet.
Examples: Stratospheric aerosol injection (sulfur), marine cloud brightening, space mirrors.
Risks: Does not address CO_2 (ocean acidification); termination shock; uneven regional effects; governance challenges.

CO_2 Removal (CDR)

Goal: Remove CO_2 from the atmosphere and store it permanently.
Examples: Afforestation, bioenergy with carbon capture and storage (BECCS), direct air capture (DAC), enhanced weathering.

Carbon Capture and Sequestration (CCS)

1. Capture CO_2 from large point sources (power plants, cement factories).
2. Compress and transport (pipeline, ship).
3. Inject into deep geological formations (depleted oil/gas reservoirs, saline aquifers).

Challenges: high cost, leakage risk (over millennia), induced seismicity (earthquakes), public acceptance.

Iron Fertilization (Ocean Geoengineering)

Concept: Add iron (as iron sulfate) to HNLC (High Nutrient, Low Chlorophyll) regions (Southern Ocean, equatorial Pacific) to stimulate phytoplankton blooms → biological pump → drawdown CO_2.
Potential side effects: harmful algal blooms, oxygen depletion, production of N_2O, possible deep-sea acidification (from respiration of extra organic matter), uncertainty about carbon sequestration efficiency (most sinking organic matter is remineralized before reaching deep ocean).

- **Biogeophysical and Biogeochemical Effects of Forests**

Effect	Mechanism	Climate impact
Biogeochemical	CO_2 uptake (carbon sink)	Cooling
Biogeophysical	Low albedo (dark surface) Evapotranspiration (latent heat)	Warming (especially in snow-covered regions) Cooling (especially in tropics)

Tropical afforestation, reforestation, and avoided deforestation provide the largest climate benefit per unit area. Temperate and boreal forests still help but their cooling effect is partially offset by albedo warming. Deciduous trees have a larger climate benefit than conifers (higher albedo in winter in snow-covered areas, and often higher evapotranspiration).

Climate-driven risks to forest carbon storage:
Droughts: tree mortality, reduced growth.
Wildfires: direct CO_2 emission, long recovery.
Insect outbreaks (bark beetles, spruce budworm): kill trees, transfer carbon to dead pools, increase fire risk.
Human disturbances (logging, fragmentation).

Non-stationarity: disturbance regimes are changing (increasing frequency, severity) under climate change.
Implication: the permanence of forest carbon sinks cannot be guaranteed. Carbon credits based on forests must account for climate-driven risks.
Key information needed for forest-based natural climate solutions (F-NCS):
Current and potential carbon storage; Risks to permanence (fire, drought, insects, humans).
Tools: remote sensing, inventory, process models, risk assessment.

- **Agriculture and Food Systems**

Agriculture is responsible for 30–35% of global greenhouse gas emissions: CO_2 (deforestation), N_2O (fertilizer, animal waste), CH_4 (cattle, rice paddies)
Freshwater withdrawals: 70% of total (irrigation). Land cover: Croplands + pastures = ~40% of ice-free land surface.
Monoculture in modern industrial farming leads to soil degradation and increased pests and diseases.
Fertilizer, animal waste and pesticide use lead to eutrophication, dead zones; air pollution.
Developing countries: Larger share due to land use change and traditional agriculture.
Industrialized countries: Larger share from livestock (CH_4, N_2O) and food processing/transport.
Sustainable agriculture management practices : (i) intercropping can reduce fertilizer use and improve crop yield; (ii) no-till farming reduces decomposition, reduces fuel use, and increases surface albedo
Irrigation can increase soil moisture, causing higher evapotranspiration and lowered Bowen ratio, which helps with surface cooling. This inhibits convection and reduce cloud cover and precipitation globally. However, the increased RH may increase CAPE, causing complex precipitation response.
Shallower PBL can trap aerosols and cause $PM_{2.5}$ to increase. High RH promotes SOA formation, while O_3 generally decreases (due to lower T and high RH, while NOx titration also helps). Water-saving methods can reverse the tradeoff in the future to mitigate O_3 impacts.
Biofuel crops (e.g., maize for ethanol, sugarcane, palm oil for biodiesel) can reduce fossil fuel CO_2, but they can compete with food production and increase fertilizer usage. Biofuel crops generally have higher albedo and evaporative cooling than food crops.
Dietary changes in China (mainly in form of higher meat consumption) have caused an increase in agricultural NH_3 emission by ~60%, annual mean $PM_{2.5}$ to rise by up to ~10 $\mu g\ m^{-3}$, out of 1.8 million PM2.5-related premature deaths in 2010, and environmental injustice – richer regions eat more meat, poorer regions suffer more air pollution.
Sustainable farming methods for production and dietary choices for consumption are important strategies to mitigate air pollution in China and environmental impacts of agriculture.